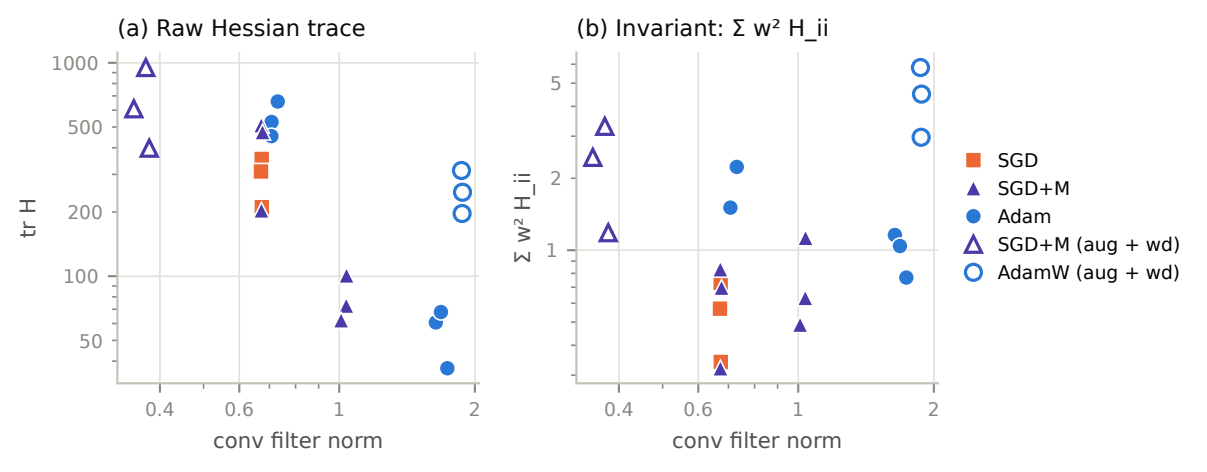
(a) Raw Hessian trace
tr H
1000
500
200
100
50
0.4
0.6
1
2
conv filter norm
(b) Invariant: Σ w² H_ii
Σ w² H_ii
5
2
1
0.4
0.6
1
2
conv filter norm
SGD
SGD+M
Adam
SGD+M (aug + wd)
AdamW (aug + wd)